

Harry Locomotive Project 3 — BT Edition

Dual Bluetooth Train Safe Distance System

Both trains BLE — cooperative speed management keeps safe gap automatically

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Track	Track 3 — outer single loop, both trains same direction
Train A	Front train — BLE controlled (update TRAIN_A_MAC in config.py)
Train B	Rear train — BLE controlled (update TRAIN_B_MAC in config.py)
Detection	Colored stickers on roofs — ceiling camera identifies both trains
Key advantage	Both BLE = emergency stops BOTH + cooperative gap management from both sides
Protocol	LionChief BLE — same as Project 1 and 2, just new MAC addresses

Before running: Open config.py and set TRAIN_A_MAC and TRAIN_B_MAC to your actual train MAC addresses. Both trains must be powered on.

HOW IT WORKS

The ceiling camera detects both trains by their colored stickers every frame and measures the pixel gap between them. Both train speeds are adjusted cooperatively — since both are BLE, the gap can be managed from both sides simultaneously for faster, smoother control.

Situation	single_track (B only)	single_trackBT (BOTH)
Gap closing	Slow Train B	Slow B + nudge A faster
Gap opening	Speed B up	Speed B up + slow A slightly
DANGER zone	Stop Train B only	STOP BOTH instantly
BLE drops on A	No effect (A free-runs)	A stops — press E to stop B too

Zone Map — Both Trains:

Zone	Gap	Train A	Train B	Action
DANGER	< 80px	STOP	STOP	Both trains stop immediately — always highest priority
WARNING	< 150px	Spd+1	Spd 2	A speeds up slightly, B slows to minimum
CAUTION	< 230px	Steady	Spd 3	A holds user speed, B reduces to caution speed
SAFE	< 320px	User	User	Both at user-set cruising speed — ideal state
FAR	> 450px	Spd-1	Max	A slows slightly, B speeds up — gap closes from both sides

COOPERATIVE_ADJUST_A in **config.py** controls how many speed steps Train A is nudged (default=1). Set to 0 to turn off Train A adjustments.

SETUP AND RUN

1 Update MACs in config.py

Open single_trackBT/config.py in VS Code

Set TRAIN_A_MAC to your front train MAC address

Set TRAIN_B_MAC to your rear train MAC address

To find MAC: power train on, scan with nRF Connect app, name starts with LC

2 Activate virtual environment and install packages

Open VS Code terminal (Ctrl+`)

→ .venv\Scripts\activate

→ pip install -r single_trackBT\requirements.txt

```
.venv\Scripts\activate  
pip install -r single_trackBT\requirements.txt
```

3 Put stickers on train roofs

Train A (front): bright RED sticker

Train B (rear): bright YELLOW sticker

At least 2x2cm, visible from ceiling camera, distinct colors

4 Calibrate colors and distances (first time)

→ cd single_trackBT

→ python calibrate.py

Click Train A sticker (TAB to switch) then click Train B sticker

Press D/W/C/S/F at each gap position, then A to save

```
cd single_trackBT  
python calibrate.py
```

5 Run the system

- Power on both trains, turn phone Bluetooth OFF
- python main.py
- Wait for both BLE symbols to show green in camera window
- Set speed with 1-7 keys — system controls gap automatically

python main.py

KEYBOARD COMMANDS

Click camera window first to enable keyboard.

1 - 7	Set BOTH trains cruising speed simultaneously
+ / -	Speed both up or down one step
E	EMERGENCY STOP — both trains immediately (use when needed urgently)
S	Normal stop — both trains
R	Resume both trains at their user-set speed
P	Pause / resume auto control
H	Horn on/off — both trains
L	Lights on/off — both trains
Q/ESC	Quit — stops both trains and disconnects cleanly

TROUBLESHOOTING

Only one train connects	Check both MACs in config.py. Is both trains powered on and in BLE range?
Train A BLE drops mid-run	Press E to emergency stop both immediately. Restart main.py.
Speeds too jerky	Reduce ALPHA_SLOW_DOWN in config.py (try 0.4). Increase MIN_COMMAND_INTERVAL_MS.
Too much Train A nudging	Set COOPERATIVE_ADJUST_A = 0 in config.py for Train B only control.
Both blobs look same	Use more distinct sticker colors. Re-run calibrate.py.
Changing a train	Update only that MAC in config.py. Everything else stays the same.
Phone BT interfering	Always turn phone Bluetooth OFF before starting main.py.

DAILY WORKFLOW

1	VS Code → Open Folder → Linchief_train
2	Ctrl+` → .venv\Scripts\activate
3	cd single_trackBT
4	Power on both trains → phone Bluetooth OFF
5	python main.py
6	Wait for both BLE ■ in camera window
7	Press 5 to set cruising speed
8	Auto control runs — trains maintain safe gap
9	Press Q when done

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